

Sharp orders of arithmetical critical groups

via canonical lattice simplices

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Abstract

We identify the critical group of an arithmetical structure on a connected simple graph with the vertex-lattice quotient of a canonical lattice simplex. Applying the sharp multiplicity theorem of Averkov, Kasprzyk, Lehmann and Nill, we determine the largest possible critical-group order among all such structures on graphs with a fixed number of vertices. Complete graphs attain the bound. The same bound holds for stars with a fixed number of leaves, proving the largest-order conjecture of Archer, Diaz-Lopez, Glass and Louwsma. We also determine the maximizing arithmetical structures on complete graphs and stars.

1 Introduction and main result

Let G be a connected simple graph on $[n] = \{1, \dots, n\}$, with $n \geq 2$, and let $A(G)$ be its adjacency matrix. An *arithmetical structure* on G is a pair of positive integer vectors (d, r) such that r is primitive and

$$Lr = 0, \quad L = \text{diag}(d) - A(G).$$

Its critical group is the finite abelian group

$$\mathcal{K}(G; d, r) = \text{Tor}(\mathbb{Z}^n / L\mathbb{Z}^n).$$

The ordinary graph Laplacian is the case $r = \mathbf{1}$. Arithmetical structures allow other diagonal entries, producing critical groups whose orders can be much larger than the numbers of spanning trees of the underlying graphs.

Write $s_1 = 2$ and $s_{j+1} = s_1 \cdots s_j + 1$ for Sylvester's sequence, and define

$$M_n = \begin{cases} n^{n-2}, & 2 \leq n \leq 4, \\ 128, & n = 5, \\ 3(s_{n-2} - 1)^2, & n \geq 6. \end{cases} \quad (1)$$

For stars we use the convention that S_n has n leaves and hence $n + 1$ vertices.

Theorem 1. *For every arithmetical structure on a connected simple graph G with $n \geq 2$ vertices,*

$$|\mathcal{K}(G; d, r)| \leq M_n.$$

For each n , the bound is attained by an arithmetical structure on the complete graph K_n . Moreover, M_n is the largest critical-group order among arithmetical structures on the star S_n .

Archer, Diaz-Lopez, Glass and Louwsma [1] determine critical groups of arithmetical structures on stars and complete graphs. Their Conjecture 24 asserts the last conclusion of Theorem 1 for $n \geq 6$, together with its complete-graph counterpart. Their notation is $a_1 = 1$ and $a_{j+1} = a_j^2 + a_j$, so that $a_j = s_j - 1$. Thus Theorem 1 proves that conjecture and gives a sharp bound over all connected simple graphs with a fixed number of vertices.

The geometric input is the sharp canonical-simplex multiplicity bound of Averkov, Kasprzyk, Lehmann and Nill [2]. We use their theorem, including its equality classification, as an external

result; we do not give a new proof of that bound. Taking the convex hull of Laplacian columns is the earlier construction of Braun and Meyer [4]. Balletti, Hibi, Meyer and Tsuchiya [3] show that ordinary Laplacian simplices of simple strongly connected digraphs have a unique interior lattice point. Our observation is that a symmetric arithmetical Laplacian gives such a simplex in the saturated lattice perpendicular to its positive kernel vector, and that its critical group is exactly the vertex-lattice quotient. This makes the multiplicity theorem applicable to the graph-order problem.

2 The simplex and its vertex-lattice quotient

For a rank- m lattice N , a full-dimensional lattice simplex $P \subset N \otimes_{\mathbb{Z}} \mathbb{R}$ is called *canonical* if its unique interior lattice point is the origin. Its *multiplicity* is the index

$$\text{mult}_N(P) = \left[N : \sum_{v \text{ a vertex of } P} \mathbb{Z}v \right].$$

In particular, the ambient lattice is part of this definition.

Lemma 2. *Let (d, r) be an arithmetical structure on a connected simple graph with $n \geq 2$ vertices. Let $v_1, \dots, v_n$ be the columns of $L = \text{diag}(d) - A(G)$ and set*

$$N = \{x \in \mathbb{Z}^n : r^\top x = 0\}, \quad P = \text{conv}(v_1, \dots, v_n).$$

Then P is a canonical $(n-1)$ -dimensional simplex in N , and

$$N / \sum_{i=1}^n \mathbb{Z}v_i \cong \mathcal{K}(G; d, r).$$

Proof. Put $R = \text{diag}(r)$. The matrix RLR is the weighted Laplacian with edge weights $r_i r_j$, because

$$r_i^2 d_i = \sum_{j: \{i,j\} \in E(G)} r_i r_j.$$

Consequently, for every $z \in \mathbb{R}^n$,

$$z^\top RLRz = \sum_{\{i,j\} \in E(G)} r_i r_j (z_i - z_j)^2.$$

Connectedness implies that L has rank $n-1$ and $\ker_{\mathbb{R}} L = \mathbb{R}r$. Symmetry puts each v_i in N . The only linear relation among the columns is a multiple of $\sum_i r_i v_i = 0$. Since $\sum_i r_i > 0$, no nonzero relation is an affine relation, so the columns are affinely independent. The positive coefficients $r_i / \sum_j r_j$ express 0 as an interior point of their simplex.

Every coordinate of every v_i is at least -1 . For each j , column v_j has positive j th coordinate d_j : every vertex has a neighbor, and $d_j r_j$ is the positive sum of the neighboring r -entries. An interior point of P uses all vertices with strictly positive coefficients. Each of its coordinates is therefore strictly greater than -1 . An interior lattice point x has all $x_j \geq 0$; the equation $r^\top x = 0$ and positivity of r force $x = 0$. This proves that P is canonical. Its vertices are primitive, since each column has an entry -1 at a neighboring vertex.

Finally, primitivity of r gives an exact sequence

$$0 \longrightarrow N/L\mathbb{Z}^n \longrightarrow \mathbb{Z}^n/L\mathbb{Z}^n \xrightarrow{r^\top} \mathbb{Z} \longrightarrow 0.$$

The left term is finite because $\text{rank } L = n - 1$. It equals the torsion subgroup of the middle term: torsion maps to zero in $\mathbb{Z}$, and every element of the finite kernel is torsion. The columns generate $L\mathbb{Z}^n$, giving the claimed isomorphism. $\square$

We state the precise geometric input in lattice language.

Theorem 3 (Averkov–Kasprzyk–Lehmann–Nill). *Let P be a canonical m -dimensional lattice simplex, where $m \geq 1$. Then*

$$\text{mult}(P) \leq M_{m+1}.$$

Up to a lattice isomorphism, the simplex attaining equality is unique. It is

$$\begin{cases} \text{conv}(0, (m+1)e_1, \dots, (m+1)e_m) - \mathbf{1}, & 1 \leq m \leq 3, \\ \text{conv}(0, 2e_1, 8e_2, 8e_3, 8e_4) - \mathbf{1}, & m = 4, \\ \text{conv}(0, s_1e_1, \dots, s_{m-2}e_{m-2}, 3te_{m-1}, 3te_m) - \mathbf{1}, & m \geq 5, \end{cases}$$

where $t = s_{m-1} - 1$ in the last line and the displayed models have ambient lattice $\mathbb{Z}^m$.

This is Proposition 2.8 and Theorem 2.9 of [2]. Combining it with Lemma 2, with $m = n - 1$, proves the upper bound in Theorem 1.

3 Attainment and the star bound

We record an elementary order formula. If L is a symmetric positive semidefinite integral matrix of rank $n - 1$ with primitive null vector r , then

$$\text{adj}(L) = crr^\top, \quad c = |\text{Tor}(\mathbb{Z}^n/L\mathbb{Z}^n)|. \quad (2)$$

Indeed the rank-one adjugate is a rational multiple of $rr^\top$. All its entries are integral, and $\gcd_{i,j}(r_i r_j) = 1$, so the multiple is an integer. The gcd of the $(n - 1)$ -minors is the order of the torsion cokernel by Smith normal form. Positivity of the nonzero eigenvalues fixes the positive sign in (2).

Let $q_1, \dots, q_n$ be positive integers with

$$\sum_{i=1}^n \frac{1}{q_i} = 1. \quad (3)$$

Put $\ell = \text{lcm}(q_1, \dots, q_n)$, $r_i = \ell/q_i$ and $d_i = q_i - 1$. All $q_i \geq 2$, and r is primitive: for each prime dividing ℓ , some denominator has its full valuation. Since $\sum_i r_i = \ell$, these data define an arithmetical structure on K_n with

$$L = \text{diag}(q_1, \dots, q_n) - J,$$

where J is the all-ones matrix. Deleting row and column j and applying the determinant lemma gives the principal cofactor

$$\prod_{i \neq j} q_i \left(1 - \sum_{i \neq j} \frac{1}{q_i} \right) = \frac{\prod_i q_i}{q_j^2}.$$

By (2),

$$|\mathcal{K}(K_n; d, r)| = \frac{\prod_i q_i}{\ell^2}. \quad (4)$$

This is also the order consequence of the group formula in [1].

For $2 \leq n \leq 4$, take $q_i = n$ for every i ; formula (4) gives $M_n = n^{n-2}$. For $n = 5$, take

$$q = (2, 8, 8, 8, 8),$$

giving order 128. For $n \geq 6$, put

$$t = s_{n-2} - 1 = \prod_{i=1}^{n-3} s_i, \quad q = (s_1, \dots, s_{n-3}, 3t, 3t, 3t). \quad (5)$$

The telescoping identity $\sum_{i=1}^{n-3} 1/s_i = 1 - 1/t$ proves (3). Here $\ell = 3t$, and (4) gives

$$\frac{t(3t)^3}{(3t)^2} = 3t^2 = M_n.$$

These are the extremal constructions already proposed in [1]; their optimality follows from the upper bound.

Now consider an arithmetical structure on S_n . Denote its center by 0, put $q_i = d_i$ at its leaves, and let d_0 be its central label. The arithmetical equations give $r_i = r_0/q_i$ and

$$\sum_{i=1}^n \frac{1}{q_i} = d_0.$$

Primitivity implies $r_0 = \ell = \text{lcm}(q_1, \dots, q_n)$. The principal cofactor obtained by deleting the center is $\prod_i q_i$, so (2) gives

$$|\mathcal{K}(S_n; d, r)| = \frac{\prod_i q_i}{\ell^2}. \quad (6)$$

Replace each q_i by $q'_i = d_0 q_i$. The new tuple satisfies (3), has least common multiple $d_0 \ell$, and determines a complete-graph structure of order

$$\frac{\prod_i q'_i}{(d_0 \ell)^2} = d_0^{n-2} |\mathcal{K}(S_n; d, r)|.$$

The complete-graph bound therefore proves the stronger inequality

$$d_0^{n-2} |\mathcal{K}(S_n; d, r)| \leq M_n. \quad (7)$$

This scaling is the order calculation underlying [1, Proposition 20]. Conversely, each maximizing unit-fraction tuple above defines a star structure with $d_0 = 1$ and order M_n by (6). This completes the proof of Theorem 1.

4 Maximizing structures on complete graphs and stars

The equality classification in the geometric theorem gives more than the numerical answer to the conjecture.

Corollary 4. *Up to permutation of vertices, the maximizing arithmetical structure on K_n is unique for each $n \geq 2$. It is given by $d_i = q_i - 1$ and $r_i = \ell/q_i$, where $\ell = \text{lcm}(q_i)$ and*

$$q = \begin{cases} (n, \dots, n), & 2 \leq n \leq 4, \\ (2, 8, 8, 8, 8), & n = 5, \\ (s_1, \dots, s_{n-3}, 3t, 3t, 3t), & n \geq 6, \end{cases}$$

with $t = s_{n-2} - 1$. For S_n with $n \geq 3$, the maximizing structure is likewise unique up to permutation of leaves: its central label is $d_0 = 1$, its leaf labels are the displayed q_i , and $(r_0, r_1, \dots, r_n) = (\ell, \ell/q_1, \dots, \ell/q_n)$.

Proof. For a complete-graph structure put $q_i = d_i + 1$ and $h = \sum_i r_i$. The equations give $q_i r_i = h$, hence $\sum_i 1/q_i = 1$. The barycentric coordinates of 0 in the simplex of Lemma 2 are $r_i/h = 1/q_i$. Lattice isomorphisms preserve these coordinates up to permutation. The equality models in Theorem 3 have reciprocal barycentric coordinates precisely the tuples in the statement. For the last model, the nonzero-axis vertices have coordinates $1/s_1, \dots, 1/s_{n-3}, 1/(3t), 1/(3t)$, and the remaining coordinate is $1/(3t)$ by the telescoping identity. The low-dimensional models give the other two tuples. Thus equality determines q up to permutation. It then determines d and the primitive vector r uniquely.

For a star with $n \geq 3$, equality in (7) forces $d_0 = 1$. Its leaf tuple therefore determines a complete-graph structure of the same order, so the first part applies. The displayed constructions establish existence in every case. $\square$

For S_2 every arithmetical critical group is trivial, and uniqueness is not asserted. Nor do we classify all graphs attaining the universal bound in Theorem 1. The numerical bound and its attainment need no such classification.

References

- [1] K. Archer, A. Diaz-Lopez, D. Glass, and J. Louwsma, *Critical groups of arithmetical structures on star graphs and complete graphs*, Electron. J. Combin. **31** (2024), no. 1, Paper P1.5. [doi:10.37236/11729](https://doi.org/10.37236/11729).
- [2] G. Averkov, A. Kasprzyk, M. Lehmann, and B. Nill, *Sharp bounds on fake weighted projective spaces with canonical singularities*, in *Varieties, Polyhedra, Computation*, EMS Series of Congress Reports, EMS Press, 2025, pp. 319–334. [doi:10.4171/ecr/22/10](https://doi.org/10.4171/ecr/22/10). Preprint: [arXiv:2105.09635](https://arxiv.org/abs/2105.09635).
- [3] G. Balletti, T. Hibi, M. Meyer, and A. Tsuchiya, *Laplacian simplices associated to digraphs*, Ark. Mat. **56** (2018), 243–264. [doi:10.4310/ARKIV.2018.v56.n2.a3](https://doi.org/10.4310/ARKIV.2018.v56.n2.a3).
- [4] B. Braun and M. Meyer, *Laplacian simplices*, Adv. Appl. Math. **114** (2020), 101976. [doi:10.1016/j.aam.2019.101976](https://doi.org/10.1016/j.aam.2019.101976).